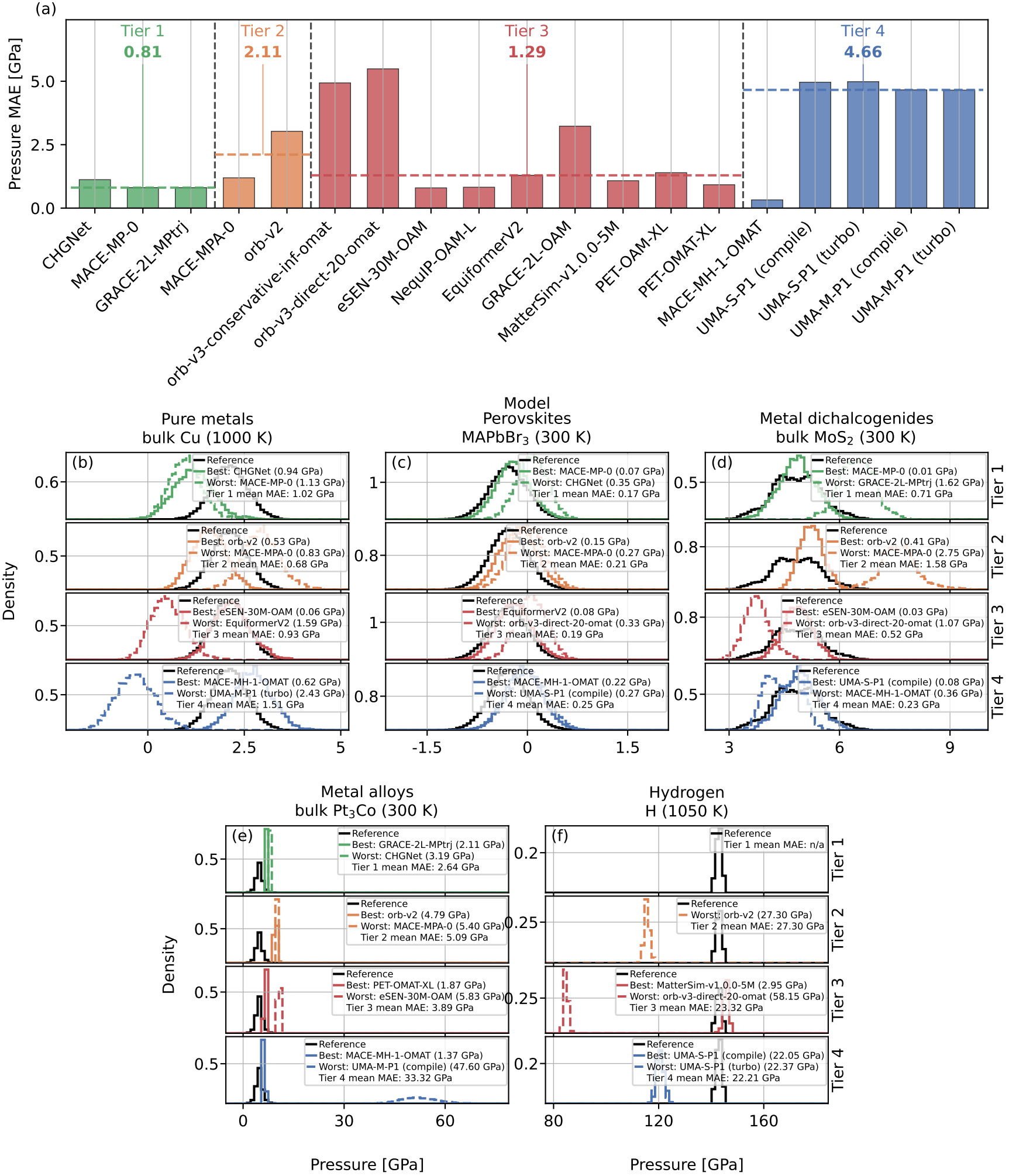
(a)
Tier 1
0.81
Tier 2
2.11
Tier 3
1.29
Tier 4
4.66
Pressure MAE [GPa]
0.0
2.5
5.0
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Pure metals
bulk Cu (1000 K)
Model
Perovskites
MAPbBr3 (300 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
(c)
(d)
Reference
Best: CHGNet (0.94 GPa)
Worst: MACE-MP-0 (1.13 GPa)
Tier 1 mean MAE: 1.02 GPa
Best: MACE-MP-0 (0.07 GPa)
Worst: CHGNet (0.35 GPa)
Tier 1 mean MAE: 0.17 GPa
Best: MACE-MP-0 (0.01 GPa)
Worst: GRACE-2L-MPtrj (1.62 GPa)
Tier 1 mean MAE: 0.71 GPa
Best: orb-v2 (0.53 GPa)
Worst: MACE-MPA-0 (0.83 GPa)
Tier 2 mean MAE: 0.68 GPa
Best: orb-v2 (0.15 GPa)
Worst: MACE-MPA-0 (0.27 GPa)
Tier 2 mean MAE: 0.21 GPa
Best: orb-v2 (0.41 GPa)
Worst: MACE-MPA-0 (2.75 GPa)
Tier 2 mean MAE: 1.58 GPa
Best: eSEN-30M-OAM (0.06 GPa)
Worst: EquiformerV2 (1.59 GPa)
Tier 3 mean MAE: 0.93 GPa
Best: EquiformerV2 (0.08 GPa)
Worst: orb-v3-direct-20-omat (0.33 GPa)
Tier 3 mean MAE: 0.19 GPa
Best: eSEN-30M-OAM (0.03 GPa)
Worst: orb-v3-direct-20-omat (1.07 GPa)
Tier 3 mean MAE: 0.52 GPa
Best: MACE-MH-1-OMAT (0.62 GPa)
Worst: UMA-M-P1 (turbo) (2.43 GPa)
Tier 4 mean MAE: 1.51 GPa
Best: MACE-MH-1-OMAT (0.22 GPa)
Worst: UMA-S-P1 (compile) (0.27 GPa)
Tier 4 mean MAE: 0.25 GPa
Best: UMA-S-P1 (compile) (0.08 GPa)
Worst: MACE-MH-1-OMAT (0.36 GPa)
Tier 4 mean MAE: 0.23 GPa
Density
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk Pt3Co (300 K)
Hydrogen
H (1050 K)
(e)
(f)
Best: GRACE-2L-MPtrj (2.11 GPa)
Worst: CHGNet (3.19 GPa)
Tier 1 mean MAE: 2.64 GPa
Tier 1 mean MAE: n/a
Best: orb-v2 (4.79 GPa)
Worst: MACE-MPA-0 (5.40 GPa)
Tier 2 mean MAE: 5.09 GPa
Worst: orb-v2 (27.30 GPa)
Tier 2 mean MAE: 27.30 GPa
Best: PET-OMAT-XL (1.87 GPa)
Worst: eSEN-30M-OAM (5.83 GPa)
Tier 3 mean MAE: 3.89 GPa
Best: MatterSim-v1.0.0-5M (2.95 GPa)
Worst: orb-v3-direct-20-omat (58.15 GPa)
Tier 3 mean MAE: 23.32 GPa
Best: MACE-MH-1-OMAT (1.37 GPa)
Worst: UMA-M-P1 (compile) (47.60 GPa)
Tier 4 mean MAE: 33.32 GPa
Best: UMA-S-P1 (compile) (22.05 GPa)
Worst: UMA-S-P1 (turbo) (22.37 GPa)
Tier 4 mean MAE: 22.21 GPa
Pressure [GPa]
Pressure [GPa]